

Experiment - 02

* AIM -> Convert the ER Diagram into a relational schema. Write complete MYSQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data demonstrate referential integrity violations.

Source Code :-

```
DROP DATABASE IF EXISTS ecommerce_db;  
CREATE DATABASE ecommerce_db;  
USE ecommerce_db;
```

-- 1. CUSTOMER (strong entity)

```
CREATE TABLE Customer (  
    customer_id INT AUTO_INCREMENT PRIMARY KEY,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) NOT NULL UNIQUE  
) ENGINE = InnoDB;
```

-- Multi valued attribute : Customer phone numbers

```
CREATE TABLE Customer-Phone (  
    customer_id INT NOT NULL,  
    phone VARCHAR(15) NOT NULL,  
    PRIMARY KEY (customer_id, phone),  
    FOREIGN KEY (customer_id) REFERENCES  
        Customer (customer_id)  
        ON DELETE CASCADE  
) ENGINE = InnoDB;
```

-- 2. ADDRESS (weak entity - depends on Customer)

```
CREATE TABLE Address (  
    customer_id INT NOT NULL,  
    address_id INT NOT NULL, ... partial key  
    street VARCHAR(100) NOT NULL,  
    city VARCHAR(50) NOT NULL,  
    state VARCHAR(50) NOT NULL,  
    pincode VARCHAR(10) NOT NULL,  
    PRIMARY KEY (customer_id, address_id)  
    FOREIGN KEY (customer_id) REFERENCE  
        Customer (customer_id) ON DELETE CASCADE  
) ENGINE = InnoDB;
```

--- 3. Category (strong entity)

```
CREATE TABLE Category (  
    category_id INT AUTO-INCREMENT, PRIMARY  
    category_name VARCHAR(50) NOTNULL UNIQUE,  
) ENGINE = InnoDB;
```

-- 4 SELECT (strong entity)
CREATE TABLE SELLER (

seller_id INT AUTO-INCREMENT PRIMARY KEY,
seller_name VARCHAR(100) NOT NULL,
email VARCHAR(100) NOT NULL UNIQUE
) ENGINE = InnoDB;

CREATE TABLE Seller-Phone (

seller_id INT NOT NULL,
phone VARCHAR(15) NOT NULL,
PRIMARY KEY (seller_id, phone),
FOREIGN KEY (seller_id) REFERENCES seller
(seller_id) ON DELETE CASCADE
) ENGINE = InnoDB;

-- 5. PRODUCT (strong entity, with specialization)
CREATE TABLE Product (

product_id INT AUTO-INCREMENT PRIMARY KEY,
product_name VARCHAR(150) NOT NULL,
price Decimal(10,2) NOT NULL,
stock INT NOT NULL DEFAULT 0,

category_id INT NULL,

seller_id INT NOT NULL,

FOREIGN KEY (category_id) REFERENCE category
(category_id)
ON DELETE SET NULL,

FOREIGN KEY (seller_id) REFERENCE seller (seller_id)
ON DELETE CASCADE

) ENGINE = InnoDB;

-- specialization: electronics (disjoint subtype of product).

```
CREATE TABLE electronics (  
    product_id INT PRIMARY KEY,  
    warranty-period NULL ... in months  
    brand VARCHAR(50),
```

```
FOREIGN KEY (product_id) REFERENCE Product  
    (product_id)
```

```
ON DELETE CASCADE
```

```
) ENGINE = InnoDB;
```

-- 6. ORDER (strong entity) - "Order" is a reserved word in MySQL

```
CREATE TABLE orders (  
    order_id INT AUTO-INCREMENT PRIMARY KEY,
```

```
order-date DATE NOT NULL,
```

```
total-amount DECIMAL(10,2) NOT NULL,
```

```
customer-id INT NOT NULL,
```

```
FOREIGN KEY (customer-id) REFERENCE customer
```

```
    (customer-id)
```

```
ON DELETE CASCADE
```

```
) ENGINE = InnoDB;
```

-- 7. Order-Item (weak entity - depends on orders)

```
CREATE TABLE Order-Item (  
    order_id INT NOT NULL,
```

```
    item-no INT NOT NULL, ... partial key
```

```
    quantity INT NOT NULL,
```

```
    unit-price DECIMAL(10,2) NOT NULL,
```

```
    product-id INT NOT NULL
```

PRIMARY KEY (order-id, item-no) -- weak entity

FOREIGN KEY (order-id) REFERENCE order (order-id)
ON DELETE CASCADE,

FOREIGN KEY (product-id) REFERENCE Product
(product-id) ON DELETE SET NULL

) ENGINE = INNODB;

-- 8. PAYMENT (~~strong~~ entity, 1:1 with orders)

CREATE TABLE Payment (

payment-id INT AUTO-INCREMENT PRIMARY,

amount DECIMAL (10,2) NOT NULL,

payment-date DATE NOT NULL,

payment-mode VARCHAR(20) NOT NULL,

order-id INT NOT NULL UNIQUE,

FOREIGN KEY (~~order-id~~) REFERENCE orders
(~~order-id~~)

ON DELETE CASCADE

) ENGINE = INNODB;

-- 9. DELIVERY (~~strong~~ entity, 1:1 with orders)

CREATE TABLE Delivery (

delivery-id INT AUTO-INCREMENT,

delivery-date DATE,

status VARCHAR(20) NOT NULL,

tracking-number VARCHAR(30) UNIQUE,

~~order-id~~ INT NOT NULL UNIQUE,

FOREIGN KEY (~~order-id~~) REFERENCE (~~orders~~)

SAMPLE DATA

INSERT INTO Customer (first-name, last-name, email)
VALUES

('Aarav', 'Sharma', 'aarav.sharma@example.com'),
('Priya', 'Iyer', 'priya.iyer@example.com');

INSERT INTO Customer-Phone (customer-id, phone)
VALUES

(1, '9876543210'),
(2, '9123456780'),
(2, '9988776655');

INSERT INTO Address (customer-id, address-id, street,
city, state, pincode) VALUES

(1, 1, '12 MG ROAD', 'LUCKNOW', 'Uttar Pradesh', '226001'),
(2, 2, '45 Anna Salai', 'Chennai', 'Tamil Nadu', '60002');

INSERT INTO CATEGORY (Category-name) VALUES
('Electronics'), ('Clothing'), ('Groceries');

INSERT INTO seller (seller-name, email) VALUES

('TechBazar Pvt Ltd', 'contact@techbazar.com'),
('FashionHub India', 'support@fashionhub.in');

INSERT INTO Seller-Phone (seller-id, phone) VALUES

(1, '011-40001234'),
(2, '022-50005678');

INSERT INTO Product (product-name, price, stock, category-
id, seller-id) VALUES

('Wireless Earbuds X200', 2499.00, 50, 1, 1),

('Cotton Kurta - blue', 899.00, 100, 2, 2);

INSERT INTO clothings (product_id, size, fabric) VALUES
(2, 'M', 'cotton');

INSERT INTO Orders (order-date, total-amount,
customer_id) VALUES
('2026-08-02', 3398.00, 1);

INSERT INTO Electronics (product_id, warranty-period,
brand) VALUES
(1, 12, 'sound wave');

INSERT INTO Order-Item (order_id, item-no, quantity,
unit-price, product_id) VALUES
(1, 1, 1, 2499.00, 1),
(1, 2, 1, 899.00, 2);

INSERT INTO Payment (amount, payment-date, payment-
mode, order-id)

VALUES (3398.00, '2026-08-01', 'UPI', 1);

INSERT INTO Delivery (delivery-date, status,
tracking-number, order-id) VALUES

('2026-08-05', 'shipped', 'TRK132465789', 1);

-- DEMONSTRATING REFERENTIAL INTEGRITY

-- (a) INSERT violation :- Order Id 999 does not exist
in orders

-- expected :- error 1452 - cannot add or update

a foreign key constraints fails (order_id FK)

-- INSERT INTO order_item (order_id, item_no, quantity, unit_price, product_id)

-- VALUES (999, 1, 2, 500.00, 1);

-- (b) INSERT violation :- product_id 999 does not exist in product

-- Expected error 1452 - foreign key constraint fails

-- INSERT INTO order_item (order_id, item_no, quantity, unit_price, product_id)

-- VALUES - (1, 3, 1, 100.00, 999);

-- (c) UNIQUE violation : this order already has a payment (1:1 constraint)

-- Expected : error 1062 - duplicate entry '1' for key 'order_id'

-- INSERT INTO Payment (amount, payment_date, payment_mode, order_id)

-- VALUES - (500.00, '2026-08-02', 'cash', 2);

-- (d) ON SET NULL demonstration :- delete a category

-- Expected :- Product category_id for its products become NULL, rows remain.

-- DELETE FROM category WHERE category_id = 1;

-- SELECT * FROM Product WHERE product_id = 1;

-- (e) ON DELETE CASCADE demonstration :-

deleted a customer

-- expected :- their Address, Customer-Phone, order rows are deleted

-- automatically, which in turn cascade to order-items,

-- Payment, and delivery for these orders

-- DELETE FROM Customer WHERE customer_id = 1;

-- SELECT * FROM orders WHERE customer_id = 1;

-- SELECT * FROM Payment WHERE Order_id = 1;

-- (f) NOT NULL violation :- order-date is required

-- expected: error 1364 - Field 'order-date' doesn't have a default value.

-- INSERT INTO orders (total-amount, customer-id)
VALUES - (500.00, 1);

17/09/20.